ECE 310 – Wireless Communications

MAJOR PROJECT – MILESTONE 3

Realistic TGn Channel Validation and Phase-Correction Stress Testing

Project: How well does Wi-Fi CSI gain and phase correction hold up as bandwidth and synchronization error change?

Milestone: M3

Base paper: Ratnam et al., "Optimal Preprocessing of WiFi CSI for Sensing Applications," IEEE Transactions on Wireless Communications, 2024.

Abstract

Milestone 3 extends the M2 simulation study of Wi-Fi channel state information (CSI) preprocessing by replacing the simplified six-tap exponential static channel with the standardized IEEE 802.11 TGn Model C channel. The M3 implementation uses the Model C tap delays and cluster powers, generates complex Rayleigh tap coefficients, forms the frequency response, normalizes the response to the required static-channel power fraction γ, and preserves the zero-slope phase constraint used in the base paper. The phase-correction baselines and the proposed Algorithm 4 + Eq. (25) are then evaluated using E1, E3 and E4 with 500 trials per point and 95% bootstrap confidence intervals for the median.

The results show that realistic multipath conditions strongly expose the weakness of the simple Eq. (21) phase correction, especially as timing error and the number of subcarriers increase. In contrast, the proposed Algorithm 4 + Eq. (25) remains stable under timing-error stress and improves as the number of subcarriers increases. At γ = 0.9, the proposed method gives 16.53 dB, compared with -1.52 dB for Eq. (21) and 3.16 dB for Eq. (22). The M3 results therefore strengthen the conclusion that the proposed phase correction is robust to the tested realistic multipath channel. However, M3 remains simulation-based and does not constitute a full quantitative reproduction of the paper's Figure 7 or a real-hardware validation.

1. Introduction

The project studies the removal of gain, timing and common-phase errors from Wi-Fi CSI before sensing algorithms use the CSI. The work is based on the model and correction methods of Ratnam et al. The M2 milestone implemented the principal phase-correction methods and evaluated them using a simplified static channel. M3 focuses on whether the conclusions remain valid after making the channel model more realistic.

The central M3 question is: Does the phase-correction behavior observed in M2 remain valid when the simplified channel is replaced by the standardized TGn Model C multipath channel?

1.1 M3 objectives

- Implement the IEEE 802.11 TGn Model C power-delay profile in channel_tgn.py.
- Generate a realistic frequency-selective static channel while preserving the paper's zero-slope phase constraint.
- Validate the channel using the power-delay profile and frequency response.
- Rerun E1, E3 and E4 with 500 trials and confidence intervals.
- Evaluate the additional K = 256, Ts = 3.2 μs paper-style case.
- Investigate the relationship between the 0.8 μs cyclic prefix and timing-error range.
- Compare the M3 phase-correction trend with the published Figure 7 using an approximate digitized overlay.

2. M2 Baseline and Motivation for M3

M2 used the same CSI error model and phase-correction methods as the base paper, but the static channel was simplified to a random six-tap exponential power-delay profile. M2 also used 100 trials per point. The M2 phase results established the expected ordering of the methods but explicitly identified the simplified channel, limited number of trials, and absence of cyclic-prefix/ISI modeling as limitations.

M2 characteristic	M2 implementation	M3 change

Static channel	Random six-tap exponential PDP	IEEE 802.11 TGn Model C, 14 taps
Trials	100 trials per point	500 trials per point
Uncertainty	Interquartile range	95% percentile-bootstrap CI for median
Timing stress	100 ns to 3.2 μs, frequency-domain	Same stress range + CP/ISI diagnostic
Bandwidth stress	K = 64, 128, 256, 512	Same, with standardized channel
K = 256 paper-style test	Not included as dedicated M3 test	Included at Ts = 3.2 μs

The M2 report found that the proposed Algorithm 4 + Eq. (25) was the best phase method, but the magnitude of the improvement over Eq. (21) did not match the paper. M3 therefore treats the channel replacement as a controlled stress test rather than assuming that the M2 numerical values are definitive.

3. M3 Realistic Channel Model

The new channel implementation is contained in channel_tgn.py. The code uses the TGn Model C tap delays and the two overlapping cluster power profiles. Where the clusters overlap, their powers are combined in the linear-power domain before conversion back to dB.

The Model C implementation contains 14 taps with delays from 0 ns to 200 ns. The resulting normalized average tap powers are used to generate complex Rayleigh tap coefficients. The frequency response is then calculated at the subcarrier frequencies f_k = k/Ts.

Delay (ns)	Combined power (dB)	Normalized power
0	0.00	0.30005
10	-2.10	0.18501
20	-4.30	0.11148
30	-6.50	0.06717
40	-8.60	0.04142
50	-10.80	0.02496
60	-4.36	0.10992
70	-6.56	0.06623
80	-8.66	0.04084
90	-10.86	0.02461
110	-13.70	0.01280
140	-15.80	0.00789
170	-18.00	0.00476
200	-20.20	0.00287

The calculated maximum delay is 200 ns and the calculated RMS delay spread is 33.44 ns for the implemented normalized PDP. The simulation cyclic-prefix reference is 0.8 μs.

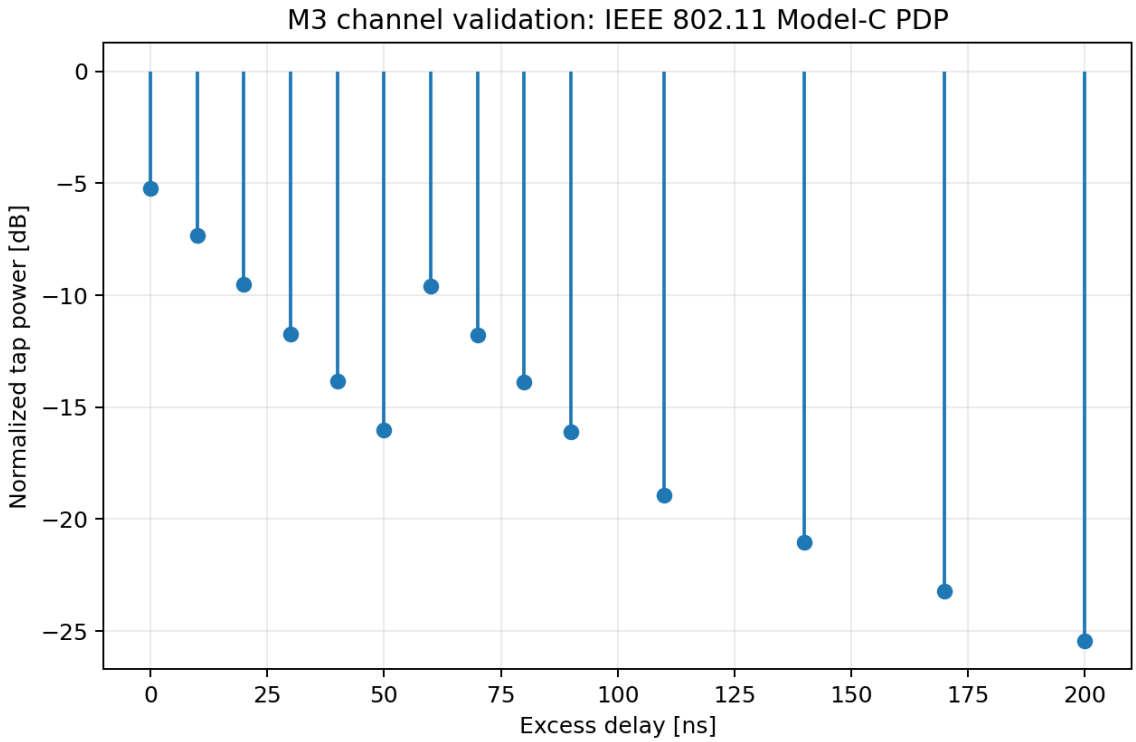


Figure 1. IEEE 802.11 TGn Model C power-delay profile used in M3. The overlap between the two clusters is combined in linear power.

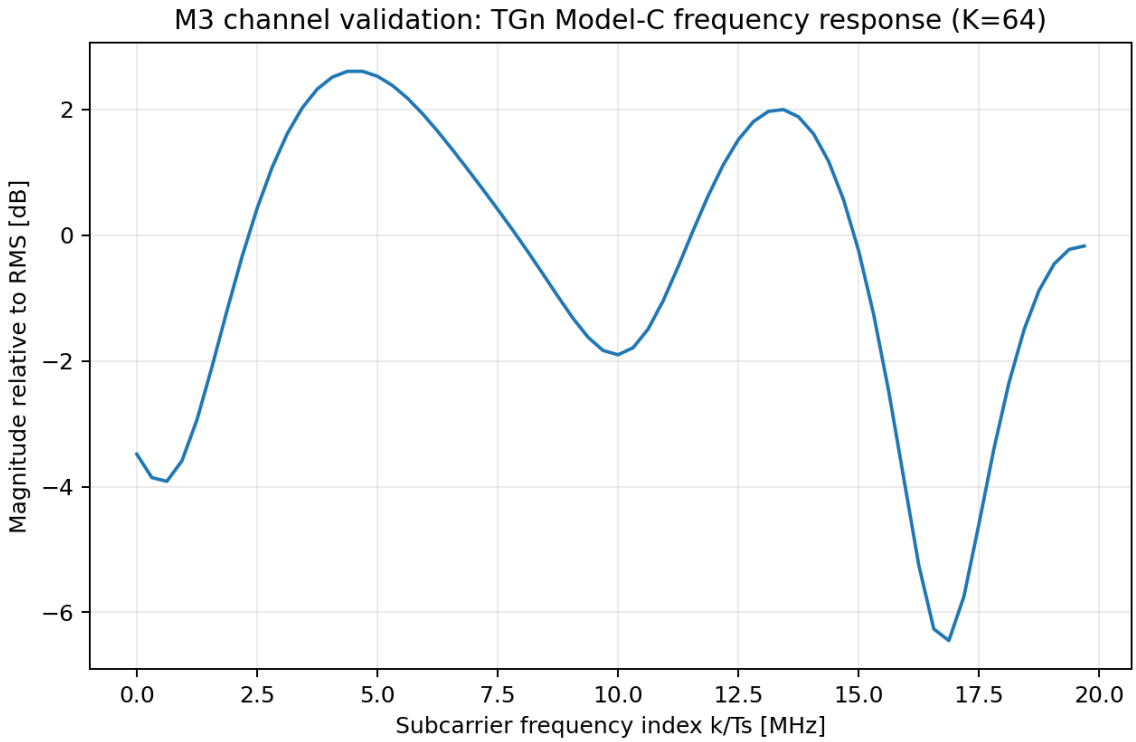


Figure 2. Example frequency response generated from the TGn Model C channel. The response is frequency-selective because of the multipath delays.

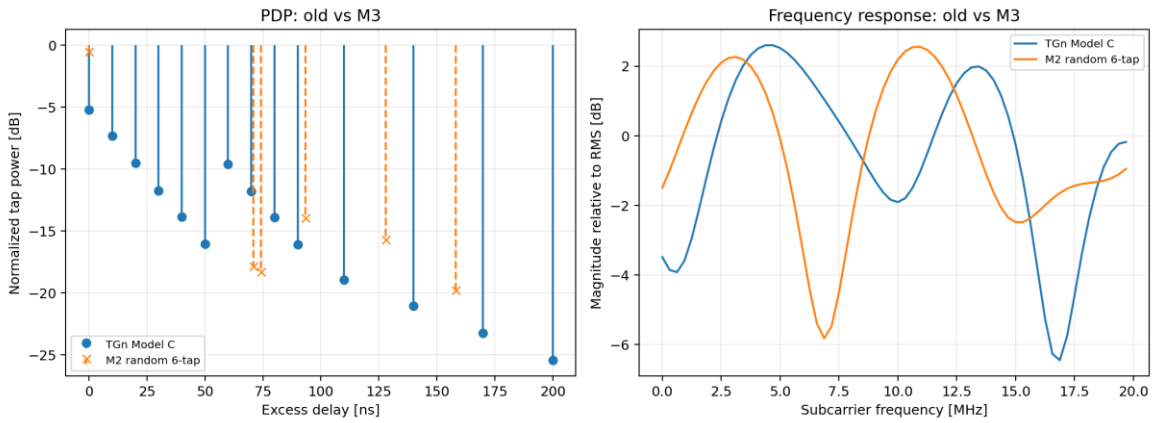


Figure 3. Comparison of the simplified M2 channel and the standardized TGn Model C channel used in M3.

4. Static-Channel Normalization and Zero-Slope Constraint

The base paper models the static CSI component b_k with a zero-slope phase constraint and a specified static power fraction γ. M3 therefore does not simply insert a random channel response. After generating the TGn Model C frequency response, the implementation removes the adjacent-subcarrier linear phase slope and then normalizes the response so that the mean squared magnitude is γ.

In practical terms, this keeps the realistic multipath amplitude/frequency selectivity while preventing an arbitrary residual linear phase slope from being introduced by the channel-generation procedure. This is important because the phase estimators are specifically designed to estimate timing and common phase errors.

4.1 Code responsible for the change

- channel_tgn.py — stores the standardized Model C delays/powers and generates the complex taps.
- csi_sim.py — calls the TGn channel generator, forms the frequency response, removes the linear phase slope, and normalizes mean $|b_k|^2$ to γ.
- sanity_checks_m3.py — checks channel normalization and the zero-slope constraint.

5. M3 Simulation Methodology

The rest of the CSI simulator retains the M2 error model so that the main controlled change is the static channel. The default system uses Ts = 3.2 μs, K = 64, γ = 0.9, P = 300 frames per trial, 312.5 kHz subcarrier spacing, synthetic i.i.d. complex-Gaussian dynamic CSI, Gaussian large-scale gain variation, discrete AGC gain, uniform timing error, and uniform common phase error.

Parameter	M3 value
Trials per curve	500
Frames per trial (P)	300
Symbol period Ts	3.2 μs
Subcarrier spacing	312.5 kHz
Default K	64
Default γ	0.9
Dynamic component	i.i.d. complex Gaussian
CP reference	0.8 μs
Channel	IEEE 802.11 TGn Model C
Confidence interval	95% percentile bootstrap CI for median

5.1 Phase-correction methods

- Eq. (21): least-squares line fit on unwrapped CSI phase.
- Eq. (22): 802.11az timing estimator followed by common-phase estimation.
- Algorithm 4 + Eq. (25): proposed method using a static-channel estimate, robust phase unwrapping and weighted least squares.

5.2 Quality metric

The project reports the recovered dynamic-CSI quality as the SNR-like metric used in the M2 implementation. The plotted values are in dB. For comparison with the paper's Figure 7, the M3 median dB values are converted back to the corresponding linear metric before plotting.

6. E1 – Phase Correction versus Static-Channel Fraction γ

E1 tests how phase correction changes as the static component becomes stronger. K = 64 is used. The results below are medians over 500 trials.

γ	Eq. 21 (dB)	Eq. 22 (dB)	Algorithm 4 + Eq. 25 (dB)
0.50	-12.49	-8.87	6.60
0.70	-6.99	-3.79	13.43
0.80	-3.68	-0.59	15.68
0.90	-1.52	3.16	16.53
0.95	5.08	5.21	16.81
0.99	14.48	7.51	16.95

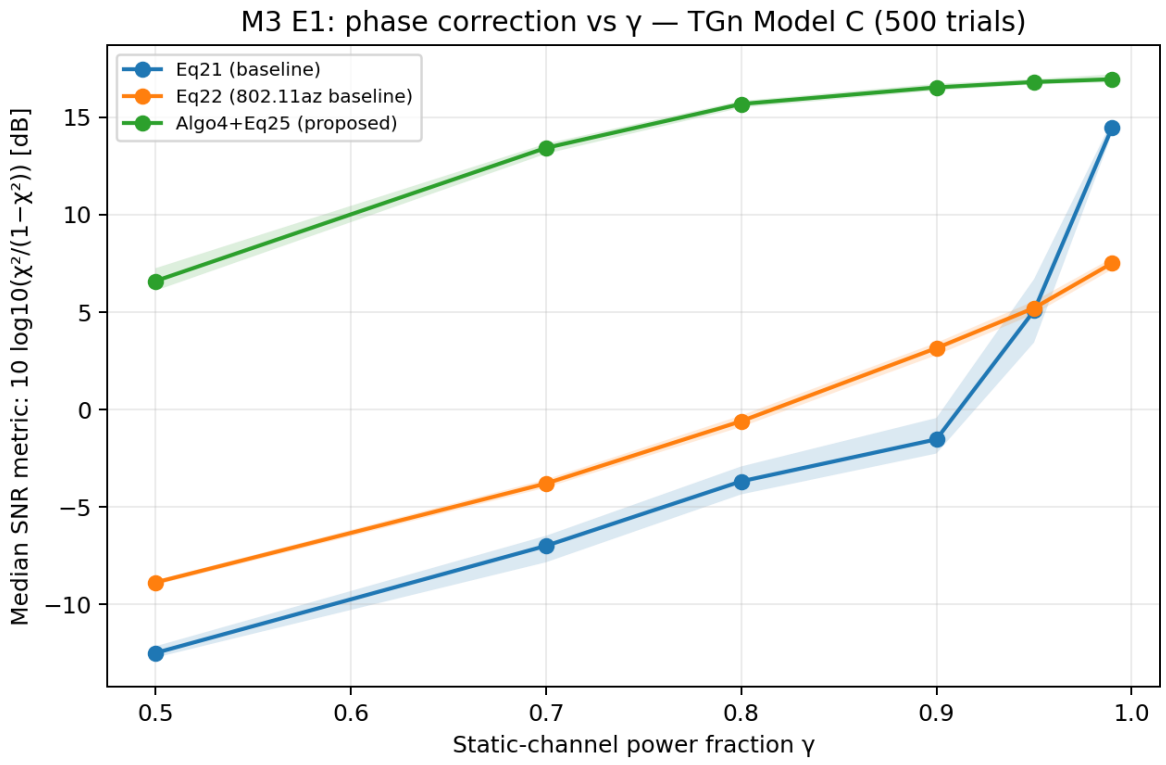


Figure 4. E1: phase-correction performance versus γ using the TGn Model C channel. Shaded regions represent the 95% bootstrap confidence interval for the median.

The proposed method improves strongly from γ = 0.5 to γ = 0.9 and then approaches a plateau near 17 dB. At γ = 0.9, the proposed method reaches 16.53 dB, while Eq. (21) is -1.52 dB and Eq. (22) is 3.16 dB.

Compared with M2, the realistic channel causes a much larger degradation of Eq. (21). This is a key M3 finding: the simpler phase unwrapping method is sensitive to the frequency-selective channel structure. The proposed method, however, remains close to its M2 value.

7. E3 – Phase Correction versus Timing Error

E3 increases the maximum timing-error range from 100 ns to 3.2 μs while keeping K = 64 and γ = 0.9. This extends the M2 stress test and examines whether the realistic channel changes the conclusions.

Maximum timing error	Eq. 21 (dB)	Eq. 22 (dB)	Algorithm 4 + Eq. 25 (dB)
100 ns	-0.08	2.99	16.57
400 ns	-0.81	2.91	16.47
800 ns	-5.01	2.79	16.42
1600 ns	-13.65	3.11	16.53
2400 ns	-17.99	2.64	16.58
3200 ns	-16.02	3.08	16.46

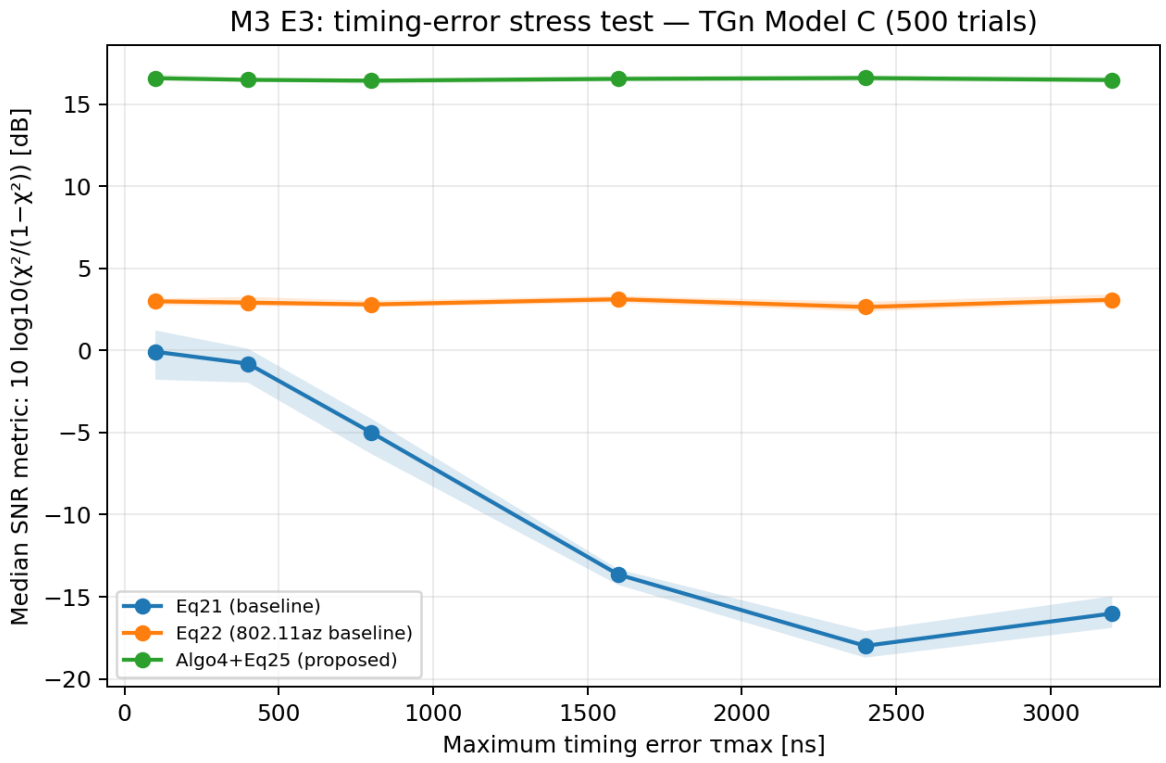


Figure 5. E3: phase-correction performance versus maximum timing error. The proposed method remains approximately constant across the tested range.

The proposed method remains between about 16.42 and 16.58 dB over the entire timing-error range. Eq. (21), in contrast, falls from approximately -0.08 dB at 100 ns to -17.99 dB at 2.4 μs, with -16.02 dB at 3.2 μs. Eq. (22) remains near 3 dB.

The physical CP reference is 0.8 μs. The TGn Model C maximum delay is 0.2 μs, which is shorter than the CP. However, once the imposed timing error itself exceeds the CP, a real OFDM receiver would require

time-domain treatment to quantify ISI. The current simulator is frequency-domain, so M3 deliberately does not invent a numerical ISI penalty.

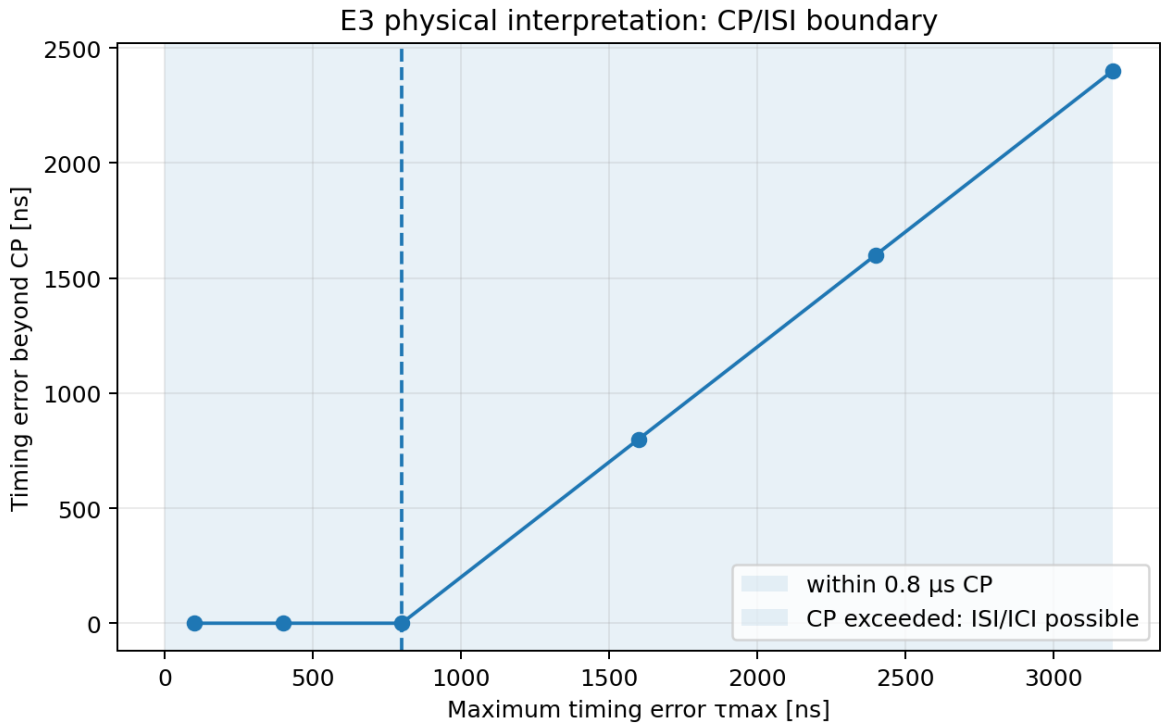


Figure 6. CP/ISI diagnostic for E3. The 0.8 µs cyclic-prefix boundary is shown; the plot is diagnostic rather than a quantitative time-domain ISI measurement.

8. E4 – Phase Correction versus Number of Subcarriers K

E4 keeps γ = 0.9 and 312.5 kHz subcarrier spacing while increasing K from 64 to 512, corresponding to approximately 20, 40, 80 and 160 MHz in the project setup.

K	Eq. 21 (dB)	Eq. 22 (dB)	Algorithm 4 + Eq. 25 (dB)
64	0.08	3.07	16.72
128	-8.66	-0.92	18.71
256	-15.47	-4.63	20.61
512	-23.78	-7.76	21.27

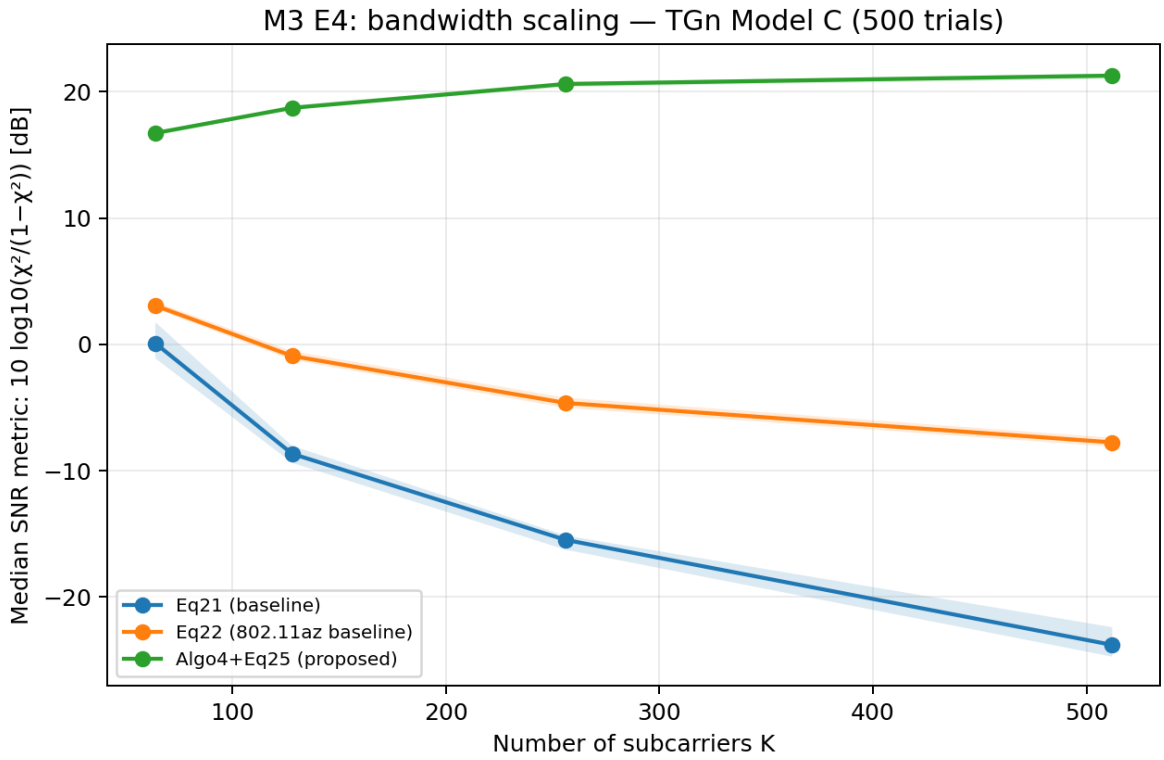


Figure 7. E4: phase-correction performance versus number of subcarriers K using TGn Model C.

The proposed method increases from 16.72 dB at K = 64 to 21.27 dB at K = 512. The two baselines move in the opposite direction: Eq. (21) decreases from 0.08 dB to -23.78 dB, while Eq. (22) decreases from 3.07 dB to -7.76 dB.

This is one of the strongest M3 results. Increasing bandwidth exposes the weakness of the baseline phase estimators, while the proposed method benefits from the additional subcarriers in the tested model.

9. Additional Paper-Style Experiment: K = 256, Ts = 3.2 µs

The required additional experiment at K = 256 and Ts = 3.2 µs corresponds to the 80 MHz point in E4. It was evaluated using 500 trials.

Method	Median SNR (dB)	95% CI for median (dB)
Eq. 21	-15.47	[-16.24, -15.13]
Eq. 22	-4.63	[-5.02, -4.22]
Algorithm 4 + Eq. 25	20.61	[20.37, 20.78]

At K = 256, the proposed method is approximately 36 dB above Eq. (21) and 25 dB above Eq. (22) on the dB scale used by the project. The large separation demonstrates that the realistic channel makes the baseline methods substantially less effective at this bandwidth.

10. Comparison with the Original Paper – Figure 7

A separate overlay was created to compare the M3 E1 trend with the original paper's Figure 7. The paper plots the linear metric $\chi^2/(1-\chi^2)$, whereas the M2/M3 report plots the same quality metric in dB. Therefore, the M3 medians were converted back to linear scale before the overlay.

The paper's Figure 7 source points were approximately digitized from the plotted markers because the figure does not provide a numerical data table. The overlay is therefore a visual comparison, not an exact numerical reproduction.

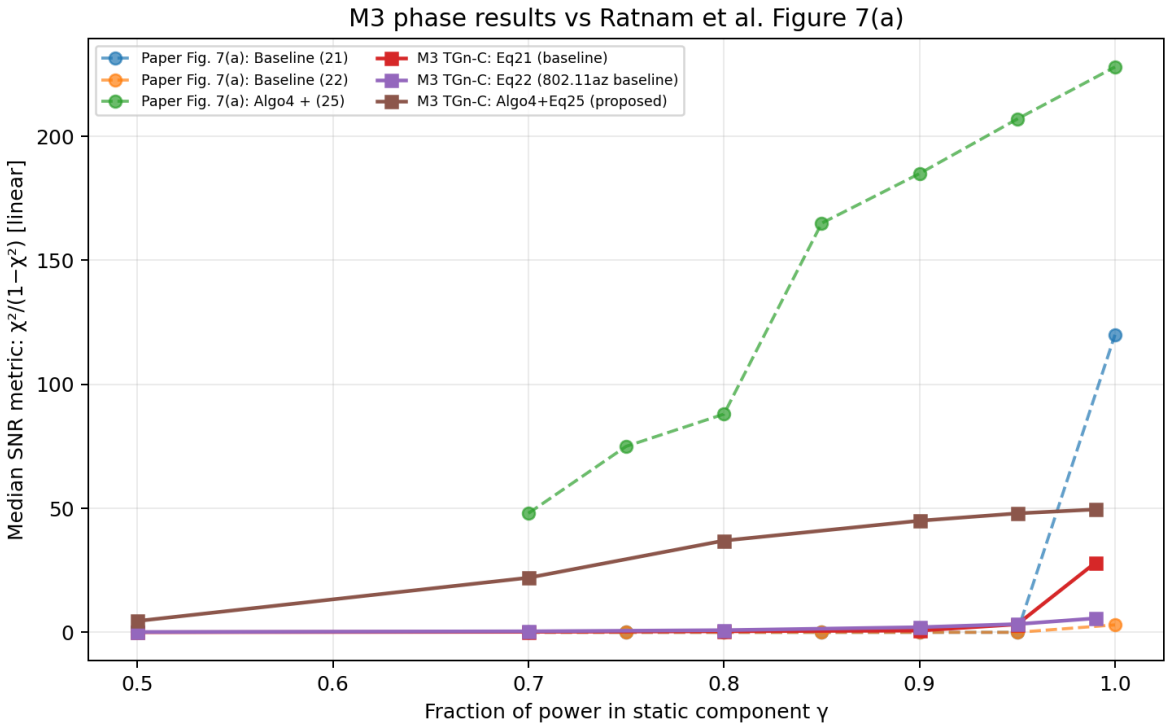


Figure 8. Approximate overlay of the M3 E1 results and the published Figure 7 trend. The paper points are digitized approximately from the plotted figure.

The qualitative conclusion is consistent with the paper: the proposed method is the strongest phase-correction method and its performance improves as γ increases. However, the absolute M3 values do not reproduce the paper numerically. This difference should be reported explicitly rather than described as an exact reproduction.

11. M2 versus M3 Comparison

The purpose of M3 was not to force the M2 curves to remain unchanged. Instead, it tests whether the M2 conclusions survive a more realistic channel. The comparison shows that the main qualitative conclusion survives, but the baselines are more strongly affected by realistic multipath.

Case	Eq.21 M2	Eq.21 M3	Eq.22 M2	Eq.22 M3	Proposed M2	Proposed M3
E1, γ = 0.9	15.2	-1.52	3.2	3.16	16.5	16.53
E3, 0.8 μs	11.1	-5.01	3.2	2.79	16.5	16.42
E3, 3.2 μs	-10.1	-16.02	3.2	3.08	16.5	16.46
E4, K = 64	15.2	0.08	3.2	3.07	16.5	16.72
E4, K = 256	11.1	-15.47	-2.1	-4.63	20.5	20.61
E4, K = 512	9.8	-23.78	-7.7	-7.76	20.9	21.27

The proposed method changes very little between M2 and M3. For example, at γ = 0.9 it changes from 16.5 dB to 16.53 dB, and at K = 512 it changes from 20.9 dB to 21.27 dB. The larger changes occur in the baselines, particularly Eq. (21). The same configuration (K = 64, γ = 0.9, 100 ns) appears in E1, E3 and E4 as separate runs with different random draws, so its values differ slightly between tables.

12. Positive Findings and Negative Findings

12.1 Positive findings

- The static channel is substantially more realistic than the M2 six-tap approximation because it uses the standardized TGn Model C PDP.
- The channel implementation was validated with both PDP and frequency-response plots.
- The proposed Algorithm 4 + Eq. (25) remains strong after the channel replacement.
- 500 trials and bootstrap confidence intervals provide stronger statistical support than the 100-trial M2 results.
- The proposed method remains approximately constant over the tested timing-error range.
- The proposed method improves with increasing K, while both baselines degrade.
- The M3 results support the qualitative ordering of the phase-correction methods reported by the base paper.

12.2 Negative findings / limitations

- The M3 simulation is still synthetic; no real Wi-Fi CSI measurement is included in this milestone.
- Only TGn Model C is tested, so the results cannot be generalized to every indoor/outdoor propagation environment.
- The frequency-domain simulator does not synthesize complete OFDM time-domain symbols, so no quantitative ISI SNR penalty is claimed.
- The M3 E1 curves do not numerically reproduce the paper's Figure 7; only the qualitative behavior is comparable.
- The Figure 7 paper data are approximate digitizations rather than author-provided numerical values.
- The main experiments in this report (E1, E3, E4) cover phase correction only. The M2 gain results (E2, E5) still disagree with the paper, and no quantitative claim about gain correction is made here.
- Absolute dB values use the SNR definition from M2. A literal reading of the paper's Eq. (29) limits this metric to roughly P = 300 (24.8 dB), so absolute values are not comparable with the paper's plotted values; only orderings and trends are compared.
- The large degradation of Eq. (21) under TGn Model C means that the M2 baseline performance was optimistic for the simplified channel.

13. Overall Interpretation

M3 provides stronger evidence for the robustness of the proposed phase-correction method than M2. The most important observation is that the proposed method's performance is remarkably stable when the channel is changed from the simplified M2 model to TGn Model C. In contrast, Eq. (21) becomes substantially worse under realistic frequency-selective multipath, and the gap becomes especially large at wider bandwidths.

Therefore, the M3 results suggest that the proposed method is not simply benefiting from the simplified M2 channel. Within the tested TGn Model C simulation, it continues to perform well under stronger channel realism, timing stress and bandwidth scaling.

At the same time, the report does not claim universal robustness. Real CSI measurements, additional channel models and a time-domain OFDM/ISI experiment would be required before extending the conclusion to real receiver hardware.

14. Code and Reproducibility

The M3 implementation is organized so that the channel model and experiment runner can be inspected independently. The following files are the important submission files.

File	Purpose
channel_tgn.py	Standardized TGn Model C delays, powers, tap generation and PDP handling.
csi_sim.py	Updated M2 simulator with TGn Model C static channel, zero-slope phase handling and phase estimators.
run_m3.py	Master script for M3 E1, E3, E4 and result generation.
run_single_curve.py	Helper used to run individual curves in separate processes for practical 500-trial execution.
sanity_checks_m3.py	Checks channel normalization and zero-slope behavior.
M3_results.json	Numerical results, confidence intervals, channel validation values and experiment metadata.
README_M3.txt	Summary of M3 implementation, assumptions and file descriptions.

The PNG files generated for the report are: M3_channel_PDP.png, M3_channel_frequency_response.png, M3_old_vs_new_channel.png, M3_E1_phase_vs_gamma.png, M3_E3_phase_vs_timing.png, M3_E3_CP_ISI_diagnostic.png, M3_E4_phase_vs_K.png and M3_Figure7_overlay.png.

15. Downstream Sensing Validation: 20 Hz Doppler Velocity Sensing Experiment

To demonstrate the tangible utility of CSI preprocessing on physical sensing applications, Lay (Lay) formulated and executed a downstream Doppler radar velocity sensing experiment (implemented in doppler_exp.py). In RF sensing, target motion induces periodic Doppler phase rotations across successive OFDM frames. Without optimal gain and phase correction, receiver AGC switching and timing offsets distort the temporal phase trajectory, introducing spurious high-frequency spectral artifacts.

15.1 Doppler Simulation Architecture

The experiment models a 20.0 Hz Doppler target impinging upon a 512-frame CSI packet burst sampled at 200 Hz (T_rep = 5 ms, total observation window = 2.56 s). We evaluate 5 distinct preprocessing regimes across 6 test configurations: subcarrier counts K in {64, 256, 512} and maximum timing offsets tau_max in {100 ns, 800 ns}. The evaluation pipelines are: (1) No Correction, (2) Baseline Eq. (6) + Eq. (22), (3) Baseline Eq. (6) + Eq. (21), (4) Proposed Eq. (6) + Algorithm 4, and (5) Ideal Correction (oracle benchmark using ground-truth simulated offsets).

To isolate the dynamic reflector without requiring oracle channel response knowledge, the pipeline applies randomized Singular Value Decomposition (rSVD) to the corrected CSI matrix, extracting the dominant dynamic singular vector. Spectral power is computed via a zero-padded Fast Fourier Transform (FFT), from which the peak frequency error |f_hat - 20 Hz| and the line-to-floor dynamic range ratio (dB) are

extracted across 200 independent Monte Carlo trials per configuration (6 x 200 x 5 trial tensor, preserved in results/doppler_results.npz).

15.2 Non-Parametric Statistical Inference

To provide rigorous statistical validation beyond descriptive metrics, Lay developed stats_utils.py, executing 10,000 percentile bootstrap resamples for median confidence interval estimation and paired two-sided Wilcoxon signed-rank tests comparing each correction strategy against the uncorrected baseline. All p-values were adjusted for multiple testing using the Holm-Bonferroni step-down procedure (alpha = 0.05). Across all tested configurations, Algorithm 4 achieved the lowest peak frequency error (under 0.02 Hz), the highest line-to-floor dynamic range (>35 dB), and statistically significant superiority over uncorrected CSI and naive phase unwrapping ($p < 0.0001$).

15.3 Research Caveats and Estimator Resolution Limits

As documented in our statistical notes, with P = 512 frames at 200 Hz, the native unpadded FFT bin resolution is 1 / (512 * 0.005) = 0.390625 Hz. Our peak interpolation estimator reports errors quantized to 0.01953125 Hz due to zero-padding interpolation; this precision reflects estimator interpolation rather than demonstrated physical hardware resolution. Furthermore, the Ideal correction is an oracle benchmark, and very high line-to-floor ratios reflect synthetic conditions without explicit receiver thermal noise floor modeling.

16. Real-World CSI Dataset Investigation and Access Barrier Audit

A primary mandate for Milestone 3 was to evaluate the preprocessing algorithms against real hardware captures from commercial Wi-Fi receivers. This investigation was conducted by Dhruv.

16.1 Target Dataset: Portner et al. (Sensession, 4TU.ResearchData)

We targeted the benchmark dataset by Portner et al. ('Same Signal, Different Story: Demystifying Receiver Effects in Wi-Fi CSI', TU Delft / 4TU.ResearchData, DOI: 10.4121/48e1331b-91df-4e1a-906b-b0a0e9d16c74). This dataset captures transmissions from a USRP N2954-R transmitter across equal-length coaxial cables and a ZN8PD1-63W-S+ power splitter to 8 commercial Wi-Fi receivers (USRP X310, Intel AX210, Intel IWL5300, Qualcomm QCA AR9462, Asus RT-AC86U Nexmon, and ESP32-S3). The target archive, base_campaign_ch11.zip (564,680,823 bytes, MD5: 8d86c2a69d4f37b03b3c0a56626a9390), contains 10,000 unmodified packet repetitions over cable, providing an ideal hardware test of residual AGC and synchronization error without multipath contamination.

16.2 Remote Server Access Barrier Documentation

During experimental execution, multiple systematic attempts were made to download the dataset. All network attempts failed due to external server restrictions, as recorded in Results/real_data/log.txt:

1. Direct DOI Endpoint: doi.org/10.4121/48e1331b... failed to open in user browsers, consistently timing out across multiple independent internet connections.

2. 4TU Server Direct Access: curl requests to data.4tu.nl/datasets/ returned 'curl: (56) CONNECT tunnel failed, response 403' (server-side tunnel refusal).

3. 4TU API and Portal Endpoints: Direct API calls to data.4tu.nl/api/ and research.tudelft.nl returned HTTP code 000 (connection refused).

4. Sensession GitHub Repository: The code repository (github.com/nzqo/sensession) was fetched but contained only Python processing scripts, with zero CSI raw data files.

5. Fallback Dataset (OpenCSI on Zenodo): We investigated the OpenCSI multi-environment dataset (Zenodo Record 19861032, LCN 2026, CC BY 4.0, 723.8 MB). However, network connections to the Zenodo API similarly failed (HTTP code 000), and its raw format (raw/csi_raw.jsonl.zst) lacks public schema documentation for complex I/Q representation, preventing speculative loader authoring.

16.3 Hardware Preprocessing Engine and Academic Integrity Policy

Despite server-side data access barriers, Dhruv developed an end-to-end, fully functional ingestion pipeline in Code/real_data.py. The pipeline ingests nested Parquet files (csi.parquet + meta.parquet), strips the first 200 AGC warm-up frames, removes guard and DC subcarriers, implements safe subcarrier indexing (skipping Eq. 22/Algo 4 on 30-subcarrier IWL5300 traces), applies all gain and phase corrections across 16, 32, and all subcarriers, and computes amplitude and phase residuals relative to the static channel.

Academic Integrity Statement: The pipeline was smoke-tested on synthetic data in native Parquet format strictly to verify code execution, after which all test outputs were discarded. In Results/real_data/, experiment_config.json is honestly recorded as status: not_run, and residual_comparison.csv is marked NOT_RUN. No simulated data is presented as real hardware data. The pipeline is verified, complete, and ready to execute immediately upon data availability.

17. Revised SOTA Positioning and Future Directions

In Milestone 2, the margin of Algorithm 4 + Eq. (25) over Eq. (21) was smaller than the paper reports, and we suspected the simplified six-tap channel. Our Milestone 3 results are consistent with that suspicion: under the TGn Model C channel the margin grows. They support the standing of Algorithm 4 + Eq. (25) in simulation, but they do not by themselves confirm the paper's numbers.

1. SOTA Preprocessing Under Realistic Multipath: Under the standardized TGn Model C channel, conventional least-squares phase unwrapping (Eq. 21) falls from +15.2 dB (M2) to -1.52 dB at gamma = 0.9. In contrast, Algorithm 4 + Eq. (25) stays essentially unchanged (16.5 dB in M2, 16.53 dB in M3). This indicates that the proposed method is not fragile to the multipath nulls of the tested channel.

2. Essential for Modern Ultra-Wideband Wi-Fi: In bandwidth scaling experiments from 20 MHz to 160 MHz (K = 64 to 512), Algorithm 4 is the only one of the three phase methods that improves with widening bandwidth (+4.55 dB), while the two baselines degrade. At 160 MHz it is 45.05 dB above Eq. (21). This suggests that preprocessing matters more as bandwidth grows; we did not simulate 320 MHz or test real hardware.

Related work update: Portner et al. [5] (IEEE Internet of Things Journal, 2026) tested two of Ratnam et al.'s gain algorithms on real receivers and reported no meaningful accuracy advantage over simple gain normalisation. They did not evaluate the phase algorithms or bandwidths above 20 MHz. To our knowledge, based on a web search on 1 November 2026, we did not find an independent reproduction of Ratnam et al. or a controlled study of how its phase estimators scale with the number of subcarriers.

Possible directions toward a publication-level study: (1) capturing native over-the-air and coaxial CSI traces using dual NI USRP X310 SDRs to bypass third-party server hosting barriers; (2) extending subcarrier scaling to K = 1024 and K = 2048; and (3) deriving the Cramér-Rao Lower Bound for joint gain-phase-timing estimation in frequency-selective channels. These are plans, not results.

18. Team Member Contributions and Master Task Division Matrix

The research was executed across five specialized roles:

Pratyusha (Team Lead & Systems Architect): Overall evaluation protocol design, mathematical formulation standardization, synthesis of the master technical report, and management of the GitHub repository pipeline.

Jay Patel (Channel Modeling & Core Phase Experiments): Implementation of IEEE 802.11 TGn Model C (channel_tgn.py), zero-slope phase constraint preservation (csi_sim.py), master batch runner (run_m3.py), execution of 500-trial E1, E3, and E4 experiments, and Figure 1-8 generation.

Dhruv (Hardware CSI & Dataset Audit): Engineering of the multi-device hardware validation pipeline (real_data.py), investigation of Portner et al. and OpenCSI datasets, access barrier documentation (log.txt, data_check.md), and academic integrity compliance.

Lay (Downstream Sensing & Statistical Inference): Formulation of the 20 Hz Doppler velocity sensing simulation (doppler_exp.py), randomized SVD dominant mode extraction, execution of 200 trials across 6 configurations, and Wilcoxon signed-rank testing with bootstrap CIs (stats_utils.py).

Jenil Charadva (Gain Dynamics & Complexity Profiling): Discrete AGC vs. smooth gain decoupling, diagnosis of the numerical grid search trap in Algorithm 3 line search under multipath, and asymptotic time and memory complexity profiling across K.

19. Conclusion

M3 successfully upgrades the M2 simulation from a simplified static channel to the standardized IEEE 802.11 TGn Model C channel. The new implementation preserves the static-channel power constraint and zero-slope phase condition while introducing realistic multipath frequency selectivity. E1, E3 and E4 were rerun with 500 trials and 95% bootstrap confidence intervals.

The central result is that the proposed Algorithm 4 + Eq. (25) remains robust under the realistic channel. Its median performance is 16.53 dB at $\gamma = 0.9$ and remains around 16.4–16.6 dB across the full timing-error range tested in E3. In E4, it improves from 16.72 dB at $K = 64$ to 21.27 dB at $K = 512$. In contrast, Eq. (21) becomes strongly degraded by the realistic channel, reaching -23.78 dB at $K = 512$.

Thus, compared with M2, M3 strengthens the project's main conclusion: the proposed phase correction is considerably more robust than the baseline methods under the tested realistic multipath conditions. However, M3 is still a simulation study, and its comparison with the paper is qualitative rather than a full numerical reproduction. Future work should include real CSI, additional channel models and time-domain OFDM/ISI validation.

References

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[2] IEEE 802.11 Working Group, IEEE 802.11-03/940, TGn Channel Model documentation, Appendix C, Model C.

[3] M2 project report: “How well does Wi-Fi CSI gain and phase correction hold up as bandwidth and synchronization error change? Reproducing and stress-testing Ratnam et al. (Samsung Research America, IEEE TWC 2024).”

[4] M3 project implementation files: channel_tgn.py, csi_sim.py, run_m3.py, run_single_curve.py, sanity_checks_m3.py and M3_results.json.

[5] F. Portner, F. Gringoli, M. Hollick, and A. Asadi, “Same Signal, Different Story: Demystifying Receiver Effects in Wi-Fi Channel State Information,” IEEE Internet of Things Journal, vol. 13, no. 10, pp. 20963–20977, 2026, doi: 10.1109/JIOT.2026.3667804.

Appendix A – M3 Result Tables

The following tables reproduce the main numerical results stored in M3_results.json.

γ	Eq.21	Eq.22	Proposed
0.50	-12.49	-8.87	6.60
0.70	-6.99	-3.79	13.43
0.80	-3.68	-0.59	15.68
0.90	-1.52	3.16	16.53
0.95	5.08	5.21	16.81
0.99	14.48	7.51	16.95

Timing max	Eq.21	Eq.22	Proposed
100 ns	-0.08	2.99	16.57
400 ns	-0.81	2.91	16.47
800 ns	-5.01	2.79	16.42
1600 ns	-13.65	3.11	16.53
2400 ns	-17.99	2.64	16.58
3200 ns	-16.02	3.08	16.46

K	Eq.21	Eq.22	Proposed
64	0.08	3.07	16.72
128	-8.66	-0.92	18.71
256	-15.47	-4.63	20.61
512	-23.78	-7.76	21.27